

Unit 6a: Earth's Atmosphere — Significance, Gases & Layers

Reading: pages 76–82 · 20 points · 15 minutes · Closed book

Name _____

Date _____

Score / 20

Answer every question. For multiple choice, circle the single best answer. Write short answers in complete sentences.

Part A · Multiple Choice

6 questions × 2 points = 12 points

1. Nitrogen and oxygen together make up about what share of Earth's atmosphere?
 - a) 78%
 - b) 88%
 - c) 99%
 - d) 100%
2. Why does air density decrease as altitude increases?
 - a) The Sun heats the upper atmosphere and drives molecules away
 - b) Ozone displaces the other gases at high altitude
 - c) The Coriolis effect throws molecules outward
 - d) Gravity pulls gas molecules toward Earth's center, and gases near sea level are compressed by the weight of the air above them
3. All of the planet's weather takes place in the
 - a) troposphere
 - b) stratosphere
 - c) mesosphere
 - d) thermosphere
4. In which layer does temperature *increase* with altitude, making the layer stable with very little mixing?
 - a) Troposphere
 - b) Stratosphere
 - c) Mesosphere
 - d) Thermosphere
5. The ozone layer sits within the stratosphere at an altitude of roughly
 - a) 5 to 10 km
 - b) 15 to 30 km
 - c) 50 to 80 km
 - d) 100 to 200 km

6. On average, temperature in the troposphere changes with altitude at a rate of about
- a) 6.5°C cooler per 1,000 m
 - b) 6.5°C cooler per 100 m
 - c) 1°C cooler per 1,000 m
 - d) 20°C cooler per 1,000 m

Part B · True or False

2 questions × 1 point = 2 points

7. During a temperature inversion, warm air sits above cold air, and the air is very stable.

TRUE / FALSE

8. The composition of the atmosphere changes dramatically as altitude increases.

TRUE / FALSE

Part C · Short Answer

2 questions × 3 points = 6 points

9. A traveler passing unprotected through the mesosphere would face severe sunburn, almost no breathable oxygen, and blood that boils at normal body temperature. Explain what causes each of these three problems.

10. What causes the aurora?

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